

Physical Observer Geometry, Protocol Holonomy, Order by Non-Closure, and Spectral Obstructions

Oscar Riveros

Abstract

A categorical agency base, a probabilistic-semantic opfibred assignment, a physical observer geometry, and a protocol category are defined. Exact coincidence of semantic profiles implies observer identity relative to separating query families. Local scientific agreement is controlled by an agreement modulus and an agreement radius. Under local continuity and a positive flattenability threshold, the agreement radius is strictly positive and finite. Abstract protocol holonomy is defined functorially; independently, realized path residue and a ball-framed path modulus are introduced, the latter yielding quantitative upper bounds on agreement radii, while local semantic dilatation yields lower bounds. Finite-bank estimators converge under a transfer modulus and an isolated threshold-crossing condition. An explicit binary system-instrument-environment model is solved in closed form, exhibiting exact semantic rigidity, nontrivial commutator residue, nontrivial protocol holonomy, and finite agreement windows. Finally, a conditional Yang–Mills reduction is formulated: exponential clustering of gauge-invariant Euclidean correlators implies a spectral mass gap, while the unresolved content is isolated into two bridge statements linking persistent UV/IR abstract holonomy and infrared curvature to exponential clustering.

1 Conventions

Throughout:

1. A denotes a small category.
2. For a pseudometric space (X, d) , the open ball convention is

$$B_r(x) := \{y \in X : d(x, y) < r\}.$$

3. If a supremum is taken over the empty subset of $[0, \infty)$, its value is declared to be 0.
4. For a measurable space X , $\mathcal{P}(X)$ denotes the set of probability measures on X .
5. d_{prot} denotes a fixed pseudometric on $\text{Ob}(\mathbf{Prot})$ (Definition 5.4). For $\pi \in \text{Ob}(\mathbf{Prot})$ and $r \geq 0$ we use the *closed* protocol ball

$$B_r^{\text{prot}}(\pi) := \{\pi' \in \text{Ob}(\mathbf{Prot}) : d_{\text{prot}}(\pi, \pi') \leq r\}.$$

6. O denotes the class of all physical observer-states of Definition 3.1.

2 Agency base and probabilistic-semantic opfibred assignment

Definition 2.1 (Agency base). An *agency base* is a small category A . Its objects are *neighborhoods of agency*. A morphism $f : a \rightarrow b$ is an *admissible transport* of local access structure from a to b .

Definition 2.2 (Local probabilistic-semantic chart). For $a \in \text{Ob}(A)$, a *local chart* is a tuple

$$U_a = (X_a, \Sigma_a, \lambda_a, G_a, \tau_a, Q_a),$$

where:

1. X_a is a standard Borel space;
2. $(X_a, \Sigma_a, \lambda_a)$ is a reference measure space;
3. $G_a \subseteq \Sigma_a$ is an accessible σ -algebra;
4. $\tau_a : X_a \rightarrow S$ is a measurable semantic map into a semantic target space S ;
5. $Q_a \subseteq L^\infty(S)$ is a family of admissible semantic queries.

Definition 2.3 (Density fibre). For $a \in \text{Ob}(A)$, define

$$D_a := \left\{ \rho \in L^1(X_a, \lambda_a) : \rho \geq 0, \int_{X_a} \rho d\lambda_a = 1 \right\}.$$

For $\rho \in D_a$, define

$$\mu_{a,\rho}(E) := \int_E \rho d\lambda_a, \quad E \in \Sigma_a.$$

Definition 2.4 (Semantic law and evaluation). For $\rho \in D_a$, define

$$\nu_{a,\rho} := (\tau_a)_\#(\mu_{a,\rho}) \in \mathcal{P}(S).$$

For $q \in Q_a$, define

$$P_{a,\rho}(q) := \int_S q(s) d\nu_{a,\rho}(s) = \int_{X_a} q(\tau_a(x)) \rho(x) d\lambda_a(x).$$

Proposition 2.5 (Uniqueness under full event equality). *Let $\rho, \eta \in D_a$. If*

$$\mu_{a,\rho}(E) = \mu_{a,\eta}(E) \quad \forall E \in \Sigma_a,$$

then $\rho = \eta$ λ_a -almost everywhere.

Proof. Equality of measures implies equality of their Radon–Nikodym derivatives with respect to λ_a . $\square$

Definition 2.6 (Observational equivalence and discrepancy). For $\rho, \eta \in D_a$, define

$$\rho \sim_a \eta \iff P_{a,\rho}(q) = P_{a,\eta}(q) \quad \forall q \in Q_a.$$

Let

$$\widehat{D}_a := D_a / \sim_a.$$

On the quotient define the *normalized discrepancy*

$$d_a([\rho]_a, [\eta]_a) := \sup_{q \in Q_a} \frac{|P_{a,\rho}(q) - P_{a,\eta}(q)|}{1 + \|q\|_\infty},$$

which is well-defined (the defining expression depends only on the classes $[\rho]_a, [\eta]_a$) and is a pseudometric on $\widehat{D}_a$. For any designated base subfamily $Q' \subseteq Q_a$ the restricted discrepancy

$$d_a^{Q'}([\rho]_a, [\eta]_a) := \sup_{q \in Q'} \frac{|P_{a,\rho}(q) - P_{a,\eta}(q)|}{1 + \|q\|_\infty}$$

is likewise well-defined on the quotient; d_a^{base} denotes the version for the designated base query family of the chart.

Definition 2.7 (Admissible transport data). Let $f : a \rightarrow b$ in A . An *admissible transport datum* over f consists of:

1. a Markov kernel K_f from X_a to X_b ;
2. a query pullback map

$$G_f : Q_b \rightarrow Q_a;$$

3. the absolute continuity condition

$$(K_f)_\#(\rho d\lambda_a) \ll \lambda_b \quad \forall \rho \in D_a;$$

4. the induced density transport

$$K_f(\rho) := \frac{d(K_f)_\#(\rho d\lambda_a)}{d\lambda_b} \in D_b;$$

5. the compatibility relation

$$P_{b,K_f(\rho)}(q) = P_{a,\rho}(G_f q) \quad \forall \rho \in D_a, \forall q \in Q_b.$$

Lemma 2.8 (Well-defined transport on observational quotients). *Let $f : a \rightarrow b$ carry admissible transport data. Then the assignment*

$$\Gamma_f : \widehat{D}_a \rightarrow \widehat{D}_b, \quad \Gamma_f([\rho]_a) := [K_f(\rho)]_b$$

is well-defined.

Proof. Assume $\rho \sim_a \eta$. Let $q \in Q_b$. By compatibility,

$$P_{b,K_f(\rho)}(q) = P_{a,\rho}(G_f q), \quad P_{b,K_f(\eta)}(q) = P_{a,\eta}(G_f q).$$

Since $G_f q \in Q_a$ and $\rho \sim_a \eta$, one has

$$P_{a,\rho}(G_f q) = P_{a,\eta}(G_f q).$$

Hence

$$P_{b,K_f(\rho)}(q) = P_{b,K_f(\eta)}(q) \quad \forall q \in Q_b,$$

so $K_f(\rho) \sim_b K_f(\eta)$. □

Definition 2.9 (Observable-state functor). Assume that for every $f : a \rightarrow b$ in A , admissible transport data are chosen so that

$$\Gamma_{\text{id}_a} = \text{id}_{\widehat{D}_a}, \quad \Gamma_{g \circ f} = \Gamma_g \circ \Gamma_f.$$

Then the assignment

$$\widehat{D} : A \rightarrow \mathbf{Set}, \quad a \mapsto \widehat{D}_a, \quad f \mapsto \Gamma_f$$

is a covariant functor.

Definition 2.10 (Category of elements and discrete opfibration). Define the *category of elements*

$$\int \widehat{D}$$

as follows:

1. objects are pairs (a, ξ) with $a \in \text{Ob}(A)$ and $\xi \in \widehat{D}_a$;
2. a morphism

$$(a, \xi) \rightarrow (b, \zeta)$$

is a morphism $f : a \rightarrow b$ in A such that

$$\Gamma_f(\xi) = \zeta.$$

The projection

$$\pi : \int \widehat{D} \rightarrow A, \quad \pi(a, \xi) = a, \quad \pi(f) = f,$$

is the canonical discrete opfibration associated with the covariant functor $\widehat{D}$ [1].

Definition 2.11 (Compatible global section). A *compatible global section* is a functor

$$s : A \rightarrow \int \widehat{D}$$

such that

$$\pi \circ s = \text{id}_A.$$

Equivalently, s assigns to each $a \in \text{Ob}(A)$ an element $s(a) \in \widehat{D}_a$ with

$$\Gamma_f(s(a)) = s(b) \quad \forall f : a \rightarrow b.$$

Theorem 2.12 (Fixed-point obstruction to compatible sections). *Assume there exists a loop*

$$a_0 \xrightarrow{f_1} a_1 \xrightarrow{f_2} \cdots \xrightarrow{f_n} a_n = a_0$$

in A , and let

$$\Gamma_\gamma := \Gamma_{f_n} \circ \cdots \circ \Gamma_{f_1} : \widehat{D}_{a_0} \rightarrow \widehat{D}_{a_0}.$$

If Γ_γ has no fixed point, then no compatible global section exists on the full subcategory generated by the loop.

Proof. If s were a compatible global section, then

$$\Gamma_\gamma(s(a_0)) = s(a_0)$$

by functoriality and $\pi \circ s = \text{id}_A$. This contradicts the absence of fixed points. $\square$

Corollary 2.13 (Relative obstruction). *Let $\xi \in \widehat{D}_{a_0}$. If $\Gamma_\gamma(\xi) \neq \xi$, then no compatible global section s on the loop-generated subcategory can satisfy $s(a_0) = \xi$.*

Proof. If $s(a_0) = \xi$, compatibility along the loop forces $\Gamma_\gamma(\xi) = \xi$. $\square$

3 Physical observer-states and exact semantic identity

Definition 3.1 (Physical observer-state). A *physical observer-state* is a tuple

$$o = (a, X_o, \Sigma_o, \lambda_o, \rho_o, G_o, \tau_o, Q_o, r_o),$$

where:

1. $a \in \text{Ob}(A)$ is a neighborhood of agency;
2. $X_o = X_S \times X_I \times X_E$ is the joint state space of system, instrument, and environment;
3. $(X_o, \Sigma_o, \lambda_o)$ is a measurable reference space;
4. ρ_o is a normalized joint density on X_o ;
5. $G_o \subseteq \Sigma_o$ is the accessible σ -algebra;
6. $\tau_o : X_o \rightarrow S$ is a semantic map;
7. $Q_o \subseteq L^\infty(S)$ is a family of admissible semantic queries;
8. r_o is the effective resource level.

Define the associated density fibre

$$D_o := \left\{ \rho \in L^1(X_o, \lambda_o) : \rho \geq 0, \int_{X_o} \rho d\lambda_o = 1 \right\},$$

so that $\rho_o \in D_o$. Define

$$\nu_o := (\tau_o)_\#(\rho_o d\lambda_o) \in \mathcal{P}(S),$$

and, for $q \in Q_o$,

$$P_o(q) := \int_S q d\nu_o = \int_{X_o} q(\tau_o(x)) \rho_o(x) d\lambda_o(x).$$

Definition 3.2 (Semantic profile). Let Q be a family of admissible semantic queries on a class $C \subseteq O$. Define

$$\Phi_Q(o) := (P_o(q))_{q \in Q}.$$

Definition 3.3 (Separating family). A family Q is *separating* on $C \subseteq O$ if

$$\Phi_Q(o) = \Phi_Q(o') \implies o = o'$$

for all $o, o' \in C$, i.e. if the profile map $o \mapsto \Phi_Q(o)$ is injective on C .

Theorem 3.4 (Exact semantic coincidence implies observer identity). *Let Q be separating on $C \subseteq O$. Then*

$$\Phi_Q(o) = \Phi_Q(o') \implies o = o'$$

for all $o, o' \in C$: exact coincidence of semantic profiles determines the observer uniquely within the class C . In particular, in the binary model of Section 8 the three-query family $\{q_0, q_1, q_\}$ is separating on D_η , and Theorem 8.1 gives the explicit inversion.*

Proof. The first assertion is the defining property of a separating family (Definition 3.3). The binary-model instance is proved in Theorem 8.1. $\square$

4 Agreement radius and non-compressibility

Definition 4.1 (Observer pseudometric and semantic discrepancy). Let d_{obs} be a pseudometric on O . Let d_{sem} be a metric or pseudometric on semantic profile space. For $\varepsilon > 0$, define

$$A_\varepsilon(o) := \{o' \in O : d_{\text{sem}}(\Phi_Q(o), \Phi_Q(o')) \leq \varepsilon\}.$$

Define the finite-distance component of o by

$$C(o) := \{o' \in O : d_{\text{obs}}(o, o') < \infty\}.$$

Definition 4.2 (Flattenability threshold). For a nonempty class $C \subseteq O$, define

$$\varepsilon_{\text{flat}}(C) := \inf \left\{ \varepsilon > 0 : \exists o^* \in C \text{ such that } \sup_{o \in C} d_{\text{sem}}(\Phi_Q(o), \Phi_Q(o^*)) \leq \varepsilon \right\}.$$

Definition 4.3 (Agreement radius). For $o \in O$, define

$$R_\varepsilon(o) := \sup \{r > 0 : B_r^{\text{obs}}(o) \subseteq A_\varepsilon(o)\}.$$

Hypothesis 4.4 (Local semantic continuity). *The profile map Φ_Q is locally continuous from (O, d_{obs}) to semantic profile space equipped with d_{sem} .*

Theorem 4.5 (Finite nonzero agreement radius). *Assume local semantic continuity. Let $o \in O$, and suppose*

$$\varepsilon < \varepsilon_{\text{flat}}(C(o)).$$

Then

$$0 < R_\varepsilon(o) < \infty.$$

Proof. By local continuity there exists $\delta > 0$ such that

$$d_{\text{obs}}(o, o') < \delta \implies d_{\text{sem}}(\Phi_Q(o), \Phi_Q(o')) \leq \varepsilon.$$

Hence $B_\delta^{\text{obs}}(o) \subseteq A_\varepsilon(o)$, so $R_\varepsilon(o) \geq \delta > 0$.

Suppose $R_\varepsilon(o) = \infty$. The defining set $\{r > 0 : B_r^{\text{obs}}(o) \subseteq A_\varepsilon(o)\}$ is downward closed, so $R_\varepsilon(o) = \infty$ forces $B_r^{\text{obs}}(o) \subseteq A_\varepsilon(o)$ for every finite $r > 0$. Hence every $o' \in C(o)$ lies in some finite ball $B_r^{\text{obs}}(o)$ (namely for any $r > d_{\text{obs}}(o, o')$), and therefore in $A_\varepsilon(o)$. Consequently

$$\sup_{o' \in C(o)} d_{\text{sem}}(\Phi_Q(o), \Phi_Q(o')) \leq \varepsilon,$$

contradicting $\varepsilon < \varepsilon_{\text{flat}}(C(o))$. □

Definition 4.6 (Collapsing compression family). Let $C \subseteq O$. A family

$$C_\lambda : C \rightarrow O, \quad \lambda \downarrow 0,$$

is *collapsing* on C if

$$\text{diam}_{d_{\text{obs}}}(C_\lambda(C)) \rightarrow 0.$$

Definition 4.7 (Semantically lossless compression). A collapsing family C_λ is *semantically lossless* on C if

$$\sup_{o \in C} d_{\text{sem}}(\Phi_Q(C_\lambda o), \Phi_Q(o)) \rightarrow 0 \quad (\lambda \downarrow 0).$$

Hypothesis 4.8 (Uniform continuity on compressed images). *For every $\varepsilon > 0$, there exists $\delta > 0$ such that for all sufficiently small λ , and all $x, y \in C_\lambda(C)$,*

$$d_{\text{obs}}(x, y) < \delta \implies d_{\text{sem}}(\Phi_Q(x), \Phi_Q(y)) < \varepsilon.$$

Theorem 4.9 (No semantically lossless collapsing compression above threshold). *Let $C \subseteq O$ with $C \neq \emptyset$ satisfy $\varepsilon_{\text{flat}}(C) > 0$. Under the uniform continuity hypothesis, there exists no family C_λ that is both collapsing and semantically lossless on C .*

Proof. Choose $0 < \varepsilon < \varepsilon_{\text{flat}}(C)/3$. By semantic losslessness, for sufficiently small λ ,

$$\sup_{o \in C} d_{\text{sem}}(\Phi_Q(C_\lambda o), \Phi_Q(o)) < \varepsilon.$$

By collapse and uniform continuity, for sufficiently small λ ,

$$\sup_{x, y \in C_\lambda(C)} d_{\text{sem}}(\Phi_Q(x), \Phi_Q(y)) < \varepsilon.$$

The two eventual conditions hold simultaneously for all sufficiently small λ ; fix such a λ , and choose $o^* \in C$. Then for all $o \in C$,

$$d_{\text{sem}}(\Phi_Q(o), \Phi_Q(o^*)) \leq d_{\text{sem}}(\Phi_Q(o), \Phi_Q(C_\lambda o)) + d_{\text{sem}}(\Phi_Q(C_\lambda o), \Phi_Q(C_\lambda o^*)) + d_{\text{sem}}(\Phi_Q(C_\lambda o^*), \Phi_Q(o^*)) < 3\varepsilon < \varepsilon_{\text{flat}}(C).$$

Hence $\sup_{o \in C} d_{\text{sem}}(\Phi_Q(o), \Phi_Q(o^*)) \leq 3\varepsilon < \varepsilon_{\text{flat}}(C)$, contradicting the definition of $\varepsilon_{\text{flat}}(C)$ as an infimum. $\square$

5 Protocol category, semantic realization, abstract holonomy, and realized path residue

5.1 Protocol category

Definition 5.1 (Universe of queries). Fix a global syntactic query universe Q^* . For every protocol state π , its admissible query family satisfies

$$Q_\pi \subseteq Q^*.$$

Definition 5.2 (Resource category). A *resource category* **Res** is a thin category whose objects are resource levels, preordered by a *scale order* $\preceq$ (writing $r \preceq r'$ when r' is a scale at least as large as r). A protocol resource r_π is an object of **Res**, and a resource update $r_\gamma : r_\pi \rightarrow r_{\pi'}$ is a morphism in **Res**.

Definition 5.3 (Protocol state). A *protocol state* is a tuple

$$\pi = (o, X_\pi, \Sigma_\pi, \lambda_\pi, \rho_\pi, Q_\pi, P_\pi, R_\pi, C_\pi, r_\pi),$$

where:

1. $o \in O$ is an observer-state;
2. $X_\pi := X_o$, $\Sigma_\pi := \Sigma_o$, $\lambda_\pi := \lambda_o$, and $\rho_\pi := \rho_o$;
3. $Q_\pi := Q_o \subseteq Q^*$;

4. P_π is a preparation rule, R_π a readout rule, and C_π a calibration, compression, or extrapolation rule;
5. $r_\pi \in \text{Ob}(\mathbf{Res})$ is a resource object. The default convention is $r_\pi := r_o$.

Definition 5.4 (Protocol category). The category **Prot** has protocol states as objects. A morphism

$$\gamma : \pi \rightarrow \pi'$$

is a triple

$$\gamma = (K_\gamma, G_\gamma, r_\gamma),$$

where:

1. $K_\gamma : \mathcal{P}(X_\pi) \rightarrow \mathcal{P}(X_{\pi'})$ is a stochastic state transport;
2. $G_\gamma : Q_{\pi'} \rightarrow Q_\pi$ is a query pullback map;
3. $r_\gamma : r_\pi \rightarrow r_{\pi'}$ is a morphism in **Res**;
4. the pullback is norm-bounded:

$$L_\gamma := \sup_{q \in Q_{\pi'}} \frac{1 + \|G_\gamma q\|_\infty}{1 + \|q\|_\infty} < \infty;$$

5. the semantic compatibility condition holds:

$$P_{\pi'}(q) = P_\pi(G_\gamma q) \quad \forall q \in Q_{\pi'}.$$

For composable $\gamma : \pi \rightarrow \pi'$ and $\gamma' : \pi' \rightarrow \pi''$, define

$$\gamma' \circ \gamma := (K_{\gamma'} \circ K_\gamma, G_\gamma \circ G_{\gamma'}, r_{\gamma'} \circ r_\gamma),$$

where the second component is the usual contravariant composition of pullback maps. The identity at π is

$$\text{id}_\pi := (\text{id}, \text{id}, \text{id}).$$

The invertible morphisms form a subgroupoid

$$\mathbf{InvProt} \subseteq \mathbf{Prot}.$$

5.2 Semantic realization and abstract holonomy

Definition 5.5 (Semantic realization category). Let **Sem** be the category whose objects are pointed pseudometric spaces

$$(S, d, s),$$

and whose morphisms are basepoint-preserving Lipschitz maps.

For each protocol state π , define

$$S_\pi := \{u : Q_\pi \rightarrow \mathbb{R} : \|u\|_\pi < \infty\}, \quad \|u\|_\pi := \sup_{q \in Q_\pi} \frac{|u(q)|}{1 + \|q\|_\infty},$$

with pseudometric

$$d_\pi(u, v) := \|u - v\|_\pi.$$

Define the distinguished semantic point

$$\Phi_\pi \in S_\pi, \quad \Phi_\pi(q) := P_\pi(q).$$

Note that $\Phi_\pi \in S_\pi$ indeed: since $P_\pi(q)$ is the expectation of q under a probability measure, $|P_\pi(q)| \leq \|q\|_\infty$, hence $\|\Phi_\pi\|_\pi \leq 1$.

Definition 5.6 (Semantic transport functor). A *semantic transport functor* is a functor

$$F : \mathbf{Prot} \rightarrow \mathbf{Sem}$$

such that, for every protocol state π ,

$$F(\pi) = (S_\pi, d_\pi, \Phi_\pi),$$

and for every morphism $\gamma : \pi \rightarrow \pi'$,

$$T_\gamma := F(\gamma) : S_\pi \rightarrow S_{\pi'}$$

is given by

$$(T_\gamma u)(q) := u(G_\gamma q), \quad q \in Q_{\pi'}.$$

Lemma 5.7 (Well-defined semantic transport). *For every protocol morphism $\gamma : \pi \rightarrow \pi'$, the map*

$$T_\gamma : S_\pi \rightarrow S_{\pi'}$$

is well-defined, Lipschitz with constant at most L_γ , and basepoint-preserving:

$$T_\gamma(\Phi_\pi) = \Phi_{\pi'}.$$

Proof. Let $u \in S_\pi$. Then for $q \in Q_{\pi'}$,

$$\frac{|(T_\gamma u)(q)|}{1 + \|q\|_\infty} = \frac{|u(G_\gamma q)|}{1 + \|q\|_\infty} \leq \|u\|_\pi \frac{1 + \|G_\gamma q\|_\infty}{1 + \|q\|_\infty}.$$

Taking the supremum over $q \in Q_{\pi'}$ yields

$$\|T_\gamma u\|_{\pi'} \leq L_\gamma \|u\|_\pi < \infty,$$

so $T_\gamma u \in S_{\pi'}$. Similarly,

$$d_{\pi'}(T_\gamma u, T_\gamma v) \leq L_\gamma d_\pi(u, v).$$

Finally, for $q \in Q_{\pi'}$,

$$(T_\gamma \Phi_\pi)(q) = \Phi_\pi(G_\gamma q) = P_\pi(G_\gamma q) = P_{\pi'}(q) = \Phi_{\pi'}(q),$$

by semantic compatibility. □

Lemma 5.8 (Functoriality of semantic transport). *For every protocol state π and every composable pair of protocol morphisms*

$$\pi \xrightarrow{\gamma} \pi' \xrightarrow{\gamma'} \pi'',$$

one has

$$T_{\text{id}_\pi} = \text{id}_{S_\pi}$$

and

$$T_{\gamma' \circ \gamma} = T_{\gamma'} \circ T_\gamma.$$

Hence the assignment F of Definition 5.6 is indeed a functor $\mathbf{Prot} \rightarrow \mathbf{Sem}$.

Proof. For $u \in S_\pi$ and $q \in Q_\pi$,

$$(T_{\text{id}_\pi} u)(q) = u(G_{\text{id}_\pi} q) = u(q),$$

so $T_{\text{id}_\pi} = \text{id}_{S_\pi}$. For composable γ, γ' and $q \in Q_{\pi''}$,

$$(T_{\gamma' \circ \gamma} u)(q) = u(G_{\gamma' \circ \gamma} q) = u((G_\gamma \circ G_{\gamma'}) q) = (T_\gamma u)(G_{\gamma'} q) = (T_{\gamma'}(T_\gamma u))(q),$$

where the middle identity uses the definition of composition in **Prot**. Thus $T_{\gamma' \circ \gamma} = T_{\gamma'} \circ T_\gamma$. $\square$

Definition 5.9 (Loop and abstract observational holonomy). A *loop* at π is a composable sequence

$$\ell = \gamma_n \circ \dots \circ \gamma_1$$

whose source and target are both π . Let

$$G_\ell := G_{\gamma_1} \circ \dots \circ G_{\gamma_n} : Q_\pi \rightarrow Q_\pi$$

be the composite pullback, so that $(T_\ell u)(q) = u(G_\ell q)$. Its *abstract observational holonomy* is the global transport defect

$$\text{Hol}_{\text{abs}}(\ell; \pi) := \sup \left\{ d_\pi(T_\ell(u), u) : u \in S_\pi, \|u\|_\pi \leq 1 \right\}.$$

By Lemma 5.7, $\text{Hol}_{\text{abs}}(\ell; \pi) \leq L_\ell + 1 \leq \prod_{i=1}^n L_{\gamma_i} + 1 < \infty$, so the holonomy is always finite. Moreover it detects exactly the non-triviality of the pullback action: if $G_\ell q = q$ for every $q \in Q_\pi$, then $T_\ell = \text{id}_{S_\pi}$ and the holonomy vanishes; conversely, if $G_\ell q \neq q$ for some $q \in Q_\pi$, then the indicator $u := \mathbf{1}_{\{G_\ell q\}}$ (the bounded functional taking value 1 at the query $G_\ell q$ and 0 elsewhere) satisfies $\|u\|_\pi = 1/(1 + \|G_\ell q\|_\infty) \leq 1$ and

$$\text{Hol}_{\text{abs}}(\ell; \pi) \geq d_\pi(T_\ell u, u) \geq \frac{|u(G_\ell q) - u(q)|}{1 + \|q\|_\infty} = \frac{1}{1 + \|q\|_\infty} > 0.$$

Thus $\text{Hol}_{\text{abs}}(\ell; \pi) > 0$ if and only if the loop's pullback acts non-trivially on the admissible queries. In contrast to the basepoint distance $d_\pi(T_\ell(\Phi_\pi), \Phi_\pi)$, which vanishes identically by Lemma 5.7, this global quantity is generally non-trivial.

Proposition 5.10 (Path-independence implies vanishing abstract holonomy). *If $T_p = T_q$ for any two composable paths $p, q : \pi \rightarrow \pi'$, then every loop at every protocol state has zero abstract observational holonomy.*

Proof. Let ℓ be a loop at π . By path-independence, the transport along ℓ agrees with the transport along the identity path at π , so

$$T_\ell = T_{\text{id}_\pi}.$$

Hence $T_\ell(u) = u$ for every $u \in S_\pi$, and

$$\text{Hol}_{\text{abs}}(\ell; \pi) = \sup_{\|u\|_\pi \leq 1} d_\pi(u, u) = 0.$$

$\square$

5.3 Realized paths and path modulus

Definition 5.11 (Realized path). A *realized path* from π_0 to π_n is a finite composable chain

$$p : \pi_0 \xrightarrow{\gamma_1} \pi_1 \xrightarrow{\gamma_2} \dots \xrightarrow{\gamma_n} \pi_n$$

in **Prot**. The path p is contained in $B_r^{\text{prot}}(\pi_0)$ (the closed protocol ball of Convention 5) if

$$\pi_i \in B_r^{\text{prot}}(\pi_0) \quad \forall i = 0, \dots, n.$$

Definition 5.12 (Realized path residue). Let

$$p : \pi_0 \xrightarrow{\gamma_1} \pi_1 \xrightarrow{\gamma_2} \dots \xrightarrow{\gamma_n} \pi_n$$

be a realized path. Define its common comparison frame by

$$Q(p) := \bigcap_{i=0}^n Q_{\pi_i} \subseteq Q^*.$$

Its *realized residue* relative to the initial state is

$$\text{Res}_{\text{obs}}(p; \pi_0) := \sup_{q \in Q(p)} \frac{|P_{\pi_n}(q) - P_{\pi_0}(q)|}{1 + \|q\|_\infty}.$$

If $Q(p) = \emptyset$, this supremum is 0 by convention. Note that for a loop ($\pi_n = \pi_0$) the realized residue vanishes identically: the residue detects the drift of the endpoint profile along non-closed paths, while the holonomy of Definition 5.9 detects the non-triviality of the pullback action even when the path closes.

Definition 5.13 (Path modulus). For a protocol state π , define

$$P_\pi(r) := \sup \left\{ \text{Res}_{\text{obs}}(p; \pi) : p \text{ is a realized path starting at } \pi, p \subseteq B_r^{\text{prot}}(\pi) \right\}.$$

Definition 5.14 (Ball-framed realized path residue). Let π be a protocol state, let $r \geq 0$, and let

$$p : \pi = \pi_0 \xrightarrow{\gamma_1} \pi_1 \xrightarrow{\gamma_2} \dots \xrightarrow{\gamma_n} \pi_n$$

be a realized path contained in $B_r^{\text{prot}}(\pi)$. Its *ball-framed realized residue* is

$$\widetilde{\text{Res}}_{\text{obs}}(p; \pi, r) := \sup_{q \in Q_{\pi, r}} \frac{|P_{\pi_n}(q) - P_\pi(q)|}{1 + \|q\|_\infty},$$

where $Q_{\pi, r}$ is the comparison frame of Definition 6.1. If $Q_{\pi, r} = \emptyset$, this supremum is 0 by convention.

Definition 5.15 (Ball-framed path modulus). For a protocol state π , define

$$\widetilde{P}_\pi(r) := \sup \left\{ \widetilde{\text{Res}}_{\text{obs}}(p; \pi, r) : p \text{ is a realized path starting at } \pi, p \subseteq B_r^{\text{prot}}(\pi) \right\}.$$

Remark 5.16 (Two path residues). The realized residue $\text{Res}_{\text{obs}}(p; \pi_0)$ measures the semantic discrepancy intrinsic to the common path frame $Q(p)$. The ball-framed residue $\widetilde{\text{Res}}_{\text{obs}}(p; \pi, r)$ is the version compatible with the agreement modulus $E_\pi(r)$, because both are measured in the same comparison frame $Q_{\pi, r}$. The unframed path modulus $P_\pi(r)$ of Definition 5.13 plays the analogous role with the common path frame $Q(p)$, and is dominated by its ball-framed counterpart whenever $Q_{\pi, r} \subseteq Q(p)$.

Theorem 5.17 (Ball-framed path modulus upper bound on agreement radius). *Let π be a protocol state, let $r > 0$, $\varepsilon > 0$, and suppose the semantic discrepancy d_{sem} of Definition 4.1 is compatible with the ball-framed comparison in the sense that*

$$d_{\text{sem}}(\Phi_Q(\pi), \Phi_Q(\pi')) \geq \sup_{q \in Q_{\pi, r}} \frac{|P_{\pi'}(q) - P_{\pi}(q)|}{1 + \|q\|_{\infty}} \quad \forall \pi' \in B_r^{\text{prot}}(\pi).$$

If

$$\tilde{P}_{\pi}(r) > \varepsilon,$$

then

$$R_{\varepsilon}(\pi) \leq r,$$

where $R_{\varepsilon}(\pi)$ is the agreement radius of Definition 4.3 taken with the closed protocol balls of Convention 5.

Proof. By the hypothesis $\tilde{P}_{\pi}(r) > \varepsilon$, there exists a realized path p starting at π , contained in $B_r^{\text{prot}}(\pi)$, whose endpoint π_n satisfies

$$\sup_{q \in Q_{\pi, r}} \frac{|P_{\pi_n}(q) - P_{\pi}(q)|}{1 + \|q\|_{\infty}} > \varepsilon.$$

By the compatibility assumption, $d_{\text{sem}}(\Phi_Q(\pi), \Phi_Q(\pi_n)) > \varepsilon$, so $\pi_n \notin A_{\varepsilon}(\pi)$, while $\pi_n \in B_r^{\text{prot}}(\pi)$. For every $s > r$ one has $B_r^{\text{prot}}(\pi) \subseteq B_s^{\text{prot}}(\pi)$, so $\pi_n \in B_s^{\text{prot}}(\pi) \setminus A_{\varepsilon}(\pi)$; hence $B_s^{\text{prot}}(\pi) \not\subseteq A_{\varepsilon}(\pi)$ for every $s > r$, and

$$R_{\varepsilon}(\pi) = \sup\{s > 0 : B_s^{\text{prot}}(\pi) \subseteq A_{\varepsilon}(\pi)\} \leq r.$$

□

The conclusion $R_{\varepsilon}(\pi) \leq r$ is sharp: strict inequality $R_{\varepsilon}(\pi) < r$ can fail. Take two protocol states π, π' with $d_{\text{prot}}(\pi, \pi') = 1$, a common query family $Q_{\pi} = Q_{\pi'} = \{q_1, q_2\}$ with $\|q_1\|_{\infty} = \|q_2\|_{\infty} = 1$, values $P_{\pi} = (q_1 : 0, q_2 : 1)$, $P_{\pi'} = (q_1 : 1, q_2 : 0)$, and a morphism $\gamma : \pi \rightarrow \pi'$ with $G_{\gamma}(q_1) = q_2$, $G_{\gamma}(q_2) = q_1$ (the compatibility condition holds and $L_{\gamma} = 1 < \infty$). Then $\tilde{P}_{\pi}(1) = \frac{1}{2}$, $E_{\pi}(r) = 0$ for $r < 1$ and $E_{\pi}(1) = \frac{1}{2}$, so for $0 < \varepsilon < \frac{1}{2}$ one has $R_{\varepsilon}(\pi) = 1 = r$: the upper bound is attained, and the strict form $R_{\varepsilon}(\pi) < r$ is false in general. (Here the compatibility hypothesis holds with d_{sem} taken as the normalized discrepancy itself.)

Definition 5.18 (Reconstructible compression loops). A family of compression morphisms $\chi_{\lambda} : \pi \rightarrow \pi_{\lambda}$ is *semantically reconstructible* if there exist reconstruction morphisms $\eta_{\lambda} : \pi_{\lambda} \rightarrow \pi$ and a modulus $\psi : [0, \infty) \rightarrow [0, \infty)$ with $\psi(t) \rightarrow 0$ as $t \downarrow 0$, such that

$$\text{Hol}_{\text{abs}}(\eta_{\lambda} \circ \chi_{\lambda}; \pi) \leq \psi \left(\sup_{q \in Q_{\pi} \cap Q_{\pi_{\lambda}}} \frac{|P_{\pi_{\lambda}}(q) - P_{\pi}(q)|}{1 + \|q\|_{\infty}} \right).$$

Theorem 5.19 (Positive compression residue forbids pointwise semantically lossless reconstructible compression). *Suppose χ_{λ} is semantically reconstructible and pointwise semantically lossless at π , i.e.*

$$\sup_{q \in Q_{\pi} \cap Q_{\pi_{\lambda}}} \frac{|P_{\pi_{\lambda}}(q) - P_{\pi}(q)|}{1 + \|q\|_{\infty}} \rightarrow 0.$$

Then

$$\text{Hol}_{\text{abs}}(\eta_{\lambda} \circ \chi_{\lambda}; \pi) \rightarrow 0.$$

Consequently, if there exists $c > 0$ and a sequence $\lambda_n \downarrow 0$ such that

$$\text{Hol}_{\text{abs}}(\eta_{\lambda_n} \circ \chi_{\lambda_n}; \pi) \geq c \quad \forall n,$$

then no such pointwise semantically lossless reconstructible compression exists.

Proof. The first statement follows immediately from the reconstructibility modulus ψ . The second is its contrapositive. $\square$

6 Agreement modulus and finite-bank invariants

Definition 6.1 (Agreement modulus). For a protocol state π and $r \geq 0$, define the common comparison frame

$$Q_{\pi,r} := \bigcap_{\pi': d_{\text{prot}}(\pi, \pi') \leq r} Q_{\pi'} \subseteq Q^*.$$

Then define

$$E_{\pi}(r) := \sup \left\{ \sup_{q \in Q_{\pi,r}} \frac{|P_{\pi'}(q) - P_{\pi}(q)|}{1 + \|q\|_{\infty}} : d_{\text{prot}}(\pi, \pi') \leq r \right\}.$$

If $Q_{\pi,r} = \emptyset$, the inner supremum is 0 by convention.

Definition 6.2 (Local overlap continuity). We say that protocol semantics is *locally overlap-continuous* at π if there exist $r_{\text{loc}} > 0$ and a nondecreasing modulus $\omega_{\pi} : [0, \infty) \rightarrow [0, \infty)$, with $\omega_{\pi}(t) \rightarrow 0$ as $t \downarrow 0$, such that for every π' with

$$d_{\text{prot}}(\pi, \pi') \leq r_{\text{loc}},$$

one has

$$\sup_{q \in Q_{\pi} \cap Q_{\pi'}} \frac{|P_{\pi'}(q) - P_{\pi}(q)|}{1 + \|q\|_{\infty}} \leq \omega_{\pi}(d_{\text{prot}}(\pi, \pi')).$$

Proposition 6.3 (Basic properties of the agreement modulus). *For every π :*

- (i) $E_{\pi}(0) = 0$ whenever $d_{\text{prot}}(\pi, \pi') = 0$ implies $\pi' = \pi$ (in particular when d_{prot} is a metric);
- (ii) if the comparison frame is constant on $[0, r^*]$, i.e. $Q_{\pi,r} = Q_{\pi,r^*}$ for all $0 \leq r \leq r^*$ (for instance when every π' with $d_{\text{prot}}(\pi, \pi') \leq r^*$ has the same query family $Q_{\pi'} = Q_{\pi}$), then

$$r_1 \leq r_2 \leq r^* \implies E_{\pi}(r_1) \leq E_{\pi}(r_2);$$

- (iii) if protocol semantics is locally overlap-continuous at π , then

$$E_{\pi}(r) \rightarrow 0 \quad (r \downarrow 0),$$

and more precisely, for $0 \leq r \leq r_{\text{loc}}$,

$$E_{\pi}(r) \leq \omega_{\pi}(r).$$

Proof. (i) If $\pi' = \pi$ is the only state at distance 0, then the closed ball $B_0^{\text{prot}}(\pi) = \{\pi\}$ and $E_\pi(0) = 0$ over the frame $Q_{\pi,0} = Q_\pi$.

(ii) With a constant frame, the inner supremum in the definition of E_π is taken over the same query family for all radii, while the outer index set $\{\pi' : d_{\text{prot}}(\pi, \pi') \leq r\}$ grows with r ; hence E_π is nondecreasing on $[0, r^*]$.

(iii) Let $0 \leq r \leq r_{\text{loc}}$ and let π' satisfy $d_{\text{prot}}(\pi, \pi') \leq r$. Since $d_{\text{prot}}(\pi, \pi) = 0$, one has

$$Q_{\pi,r} \subseteq Q_\pi \cap Q_{\pi'},$$

and therefore

$$\sup_{q \in Q_{\pi,r}} \frac{|P_{\pi'}(q) - P_\pi(q)|}{1 + \|q\|_\infty} \leq \sup_{q \in Q_\pi \cap Q_{\pi'}} \frac{|P_{\pi'}(q) - P_\pi(q)|}{1 + \|q\|_\infty} \leq \omega_\pi(d_{\text{prot}}(\pi, \pi')) \leq \omega_\pi(r).$$

Taking the outer supremum over all such π' yields

$$E_\pi(r) \leq \omega_\pi(r).$$

Since $\omega_\pi(r) \rightarrow 0$ as $r \downarrow 0$, the conclusion follows. $\square$

Definition 6.4 (Agreement radius as inverse modulus).

$$R_\varepsilon(\pi) := \sup\{r > 0 : E_\pi(r) \leq \varepsilon\}.$$

This definition is the protocol-level instance of Definition 4.3: whenever the semantic discrepancy is the normalized comparison, i.e.

$$d_{\text{sem}}(\Phi_Q(\pi), \Phi_Q(\pi')) = \sup_{q \in Q_\pi \cap Q_{\pi'}} \frac{|P_{\pi'}(q) - P_\pi(q)|}{1 + \|q\|_\infty},$$

the condition $E_\pi(r) \leq \varepsilon$ is equivalent to $B_r^{\text{prot}}(\pi) \subseteq A_\varepsilon(\pi)$ whenever the comparison frames are maximal, i.e. $Q_{\pi,r} = Q_\pi \cap Q_{\pi'}$ for every π' with $d_{\text{prot}}(\pi, \pi') \leq r$ (which holds, for instance, when every π' in the ball shares the query family $Q_{\pi'} = Q_\pi$); under that condition the two definitions of $R_\varepsilon(\pi)$ agree. In the binary model of Section 8 all queries have sup-norm 1, so the normalized modulus equals half of the profile discrepancy; Section 8 accordingly works with Definition 4.3 in d_{sem} -units, and the two conventions differ only by the constant factor 2.

Definition 6.5 (Upper semantic dilatation).

$$\text{Dil}_Q^+(\pi) := \limsup_{r \downarrow 0} \frac{E_\pi(r)}{r}.$$

The upper dilatation is the optimal local Lipschitz constant of the agreement modulus. Indeed, if $\text{Dil}_Q^+(\pi) < \infty$, then for every $L > \text{Dil}_Q^+(\pi)$ there exists $r_0 > 0$ with $E_\pi(r) \leq Lr$ on $[0, r_0]$, so by Proposition 6.6 below,

$$R_\varepsilon(\pi) \geq \min\left(r_0, \frac{\varepsilon}{L}\right),$$

If, in addition, the ratio attains its limsup locally, i.e. there exist $r_1 > 0$ with

$$E_\pi(r) \leq \text{Dil}_Q^+(\pi) r \quad (0 \leq r \leq r_1)$$

(as in the binary model of Section 8, where for interior states the linear bound $E_\pi(r) \leq r$ holds from $r = 0$, with equality attained), then Proposition 6.6 with $L_\pi = \text{Dil}_Q^+(\pi)$ gives

$$R_\varepsilon(\pi) \geq \min\left(r_1, \frac{\varepsilon}{\text{Dil}_Q^+(\pi)}\right).$$

Thus the local semantic dilatation controls the agreement radius from below, as announced in the abstract.

Proposition 6.6 (Local lower bound). *If there exist $L_\pi < \infty$ and $r_0 > 0$ such that*

$$E_\pi(r) \leq L_\pi r \quad (0 \leq r \leq r_0),$$

then

$$R_\varepsilon(\pi) \geq \min\left(r_0, \frac{\varepsilon}{L_\pi}\right).$$

Proof. If $r \leq r_0$ and $r \leq \varepsilon/L_\pi$, then $E_\pi(r) \leq \varepsilon$. □

Definition 6.7 (Finite protocol bank). A *finite protocol bank* rooted at π is a finite set $B_N(\pi) \subseteq \text{Ob}(\mathbf{Prot})$ such that $\pi \in B_N(\pi)$.

Definition 6.8 (Bank-admissible finite query family). Let $B_N(\pi)$ be a rooted finite protocol bank. A finite family

$$Q_N(\pi)$$

is *bank-admissible* if

$$Q_N(\pi) \subseteq \bigcap_{\pi' \in B_N(\pi)} Q_{\pi'} \subseteq Q^*.$$

It is *bank-separating* if the map

$$\pi' \longmapsto (P_{\pi'}(q))_{q \in Q_N(\pi)}$$

is injective on $B_N(\pi)$.

Definition 6.9 (Finite-bank estimator). Let $B_N(\pi)$ be a rooted finite protocol bank and let $Q_N(\pi)$ be a bank-admissible finite query family. Define

$$\widehat{d}_{\text{sem},N}(\pi, \pi') := \sup_{q \in Q_N(\pi)} |P_\pi(q) - P_{\pi'}(q)|,$$

$$\widehat{E}_{\pi,N}(r) := \max\left\{\widehat{d}_{\text{sem},N}(\pi, \pi') : \pi' \in B_N(\pi), d_{\text{prot}}(\pi, \pi') \leq r\right\},$$

$$\widehat{R}_{\varepsilon,N}(\pi) := \sup\{r > 0 : \widehat{E}_{\pi,N}(r) \leq \varepsilon\}.$$

Remark 6.10 (Normalization convention for the finite estimator). The continuous modulus $E_\pi(r)$ is normalized by the factor $(1 + \|q\|_\infty)^{-1}$, whereas the finite-bank estimator $\widehat{d}_{\text{sem},N}$ is left unnormalized. This asymmetry is intentional: Hypothesis 6.11 is understood to absorb the resulting comparison constants between the finite bank and the continuous modulus on each fixed window.

Hypothesis 6.11 (Transfer modulus). *There exists $\omega_N \rightarrow 0$ such that for every finite window $0 \leq r \leq r^*$,*

$$\sup_{0 \leq r \leq r^*} |\widehat{E}_{\pi,N}(r) - E_\pi(r)| \leq \omega_N.$$

Definition 6.12 (Isolated threshold crossing). Fix $\varepsilon > 0$. We say that E_π has an *isolated threshold crossing* at $R_\varepsilon(\pi)$ if there exists $\delta_0 > 0$ such that for every $0 < \delta < \delta_0$, whenever $R_\varepsilon(\pi) \pm \delta$ lie in the window under consideration,

$$E_\pi(R_\varepsilon(\pi) - \delta) < \varepsilon$$

and

$$E_\pi(R_\varepsilon(\pi) + \delta) > \varepsilon.$$

Theorem 6.13 (Consistency of empirical agreement radius). *Assume the transfer modulus hypothesis (Hypothesis 6.11) on the window $[0, r^*]$, and an isolated threshold crossing at $R_\varepsilon(\pi)$ with $0 < R_\varepsilon(\pi) < r^*$. Then*

$$\widehat{R}_{\varepsilon, N}(\pi) \rightarrow R_\varepsilon(\pi) \quad (N \rightarrow \infty).$$

Proof. Since the query family $Q_N(\pi)$ is fixed, the map $r \mapsto \widehat{E}_{\pi, N}(r)$ is nondecreasing (its index set $\{\pi' \in B_N(\pi) : d_{\text{prot}}(\pi, \pi') \leq r\}$ grows with r).

Fix $\delta > 0$ with $\delta < \min(\delta_0, R_\varepsilon(\pi), r^* - R_\varepsilon(\pi))$. By isolated crossing,

$$E_\pi(R_\varepsilon(\pi) - \delta) < \varepsilon < E_\pi(R_\varepsilon(\pi) + \delta).$$

Let

$$\alpha_\delta := \varepsilon - E_\pi(R_\varepsilon(\pi) - \delta) > 0, \quad \beta_\delta := E_\pi(R_\varepsilon(\pi) + \delta) - \varepsilon > 0.$$

Choose N so that $\omega_N < \min(\alpha_\delta, \beta_\delta)$. Then

$$\widehat{E}_{\pi, N}(R_\varepsilon(\pi) - \delta) \leq E_\pi(R_\varepsilon(\pi) - \delta) + \omega_N < \varepsilon,$$

and

$$\widehat{E}_{\pi, N}(R_\varepsilon(\pi) + \delta) \geq E_\pi(R_\varepsilon(\pi) + \delta) - \omega_N > \varepsilon.$$

Hence $R_\varepsilon(\pi) - \delta \in \{r > 0 : \widehat{E}_{\pi, N}(r) \leq \varepsilon\}$, so $\widehat{R}_{\varepsilon, N}(\pi) \geq R_\varepsilon(\pi) - \delta$; and by monotonicity of $\widehat{E}_{\pi, N}$, the inequality $\widehat{E}_{\pi, N}(R_\varepsilon(\pi) + \delta) > \varepsilon$ forces $\widehat{R}_{\varepsilon, N}(\pi) \leq R_\varepsilon(\pi) + \delta$. Therefore

$$R_\varepsilon(\pi) - \delta \leq \widehat{R}_{\varepsilon, N}(\pi) \leq R_\varepsilon(\pi) + \delta.$$

Since $\delta > 0$ is arbitrary, the conclusion follows. $\square$

7 Order induced by non-closure

Definition 7.1 (Obstruction vector). Let Z be a class of semantic states, realized as a class of protocol states (Section 5). Write a protocol state $z \in Z$ as $z = (a(z), \rho_z, \dots)$ with agency neighborhood $a(z) \in \text{Ob}(A)$ and density ρ_z , and set $\xi_z := [\rho_z]_{a(z)} \in \widehat{D}_{a(z)}$. For each resource level r , define five nonnegative obstruction functionals on Z :

(1) *Certification curvature floor.*

$$\kappa_r(z) := \inf \left\{ \text{err}(\sigma, z) : \sigma \in \Sigma_{\text{cert}}, \text{cost}(\sigma) \leq r \right\},$$

where Σ_{cert} , cost , and err are the certification data of Section 10 (Definitions 10.3 and 10.4); if no certificate of cost at most r exists, or if $z \notin \mathcal{C}_{IR}^{g.i.}$, set $\kappa_r(z) := 0$.

(2) *Atlas holonomy.*

$$\text{Hol}_r^{\text{atlas}}(z) := \sup \left\{ d_{a(z)}^{\text{base}}(\Gamma_\gamma(\xi_z), \xi_z) : \gamma \text{ a loop at } a(z) \text{ in } A \right\},$$

the maximal displacement of the state's observational class under the loop transports of the agency base, measured by the base-query discrepancy $d_{a(z)}^{\text{base}}$ of Definition 2.6.

(3) *Protocol holonomy.*

$$\text{Hol}_r^{\text{prot}}(z) := \sup \left\{ \text{Hol}_{\text{abs}}(\ell; z) : \ell \text{ a loop at } z \text{ contained in } B_r^{\text{prot}}(z) \right\}.$$

(4) *Realized (path) residue.*

$$g_r(z) := \sup \left\{ \text{Res}_{\text{obs}}(p; z) : p \text{ a realized path starting at } z \text{ and contained in } B_r^{\text{prot}}(z) \right\}.$$

(5) *Compression obstruction.*

$$c_r(z) := \varepsilon_{\text{flat}}(C_r(z)), \quad C_r(z) := \{z' \in Z : d_{\text{prot}}(z, z') \leq r\},$$

the flattenability threshold of the scale- r neighborhood of z .

Define the *obstruction vector*

$$\Delta_r(z) = \left(\kappa_r(z), \text{Hol}_r^{\text{atlas}}(z), \text{Hol}_r^{\text{prot}}(z), g_r(z), c_r(z) \right) \in [0, \infty)^5.$$

The five functionals connect the machinery of the preceding sections: the atlas holonomy $\text{Hol}_r^{\text{atlas}}$ is the fixed-point obstruction of Theorem 2.12 measured quantitatively by the discrepancy of Definition 2.6; the protocol holonomy $\text{Hol}_r^{\text{prot}}$ is the loop defect of Definition 5.9; the realized residue g_r is controlled by the ball-framed modulus and, by Theorem 5.17, $g_r(z) > \varepsilon$ implies $R_\varepsilon(z) \leq r$; the compression obstruction c_r is exactly the threshold that Theorem 4.9 shows to be incompressible; and the curvature floor κ_r aggregates to the infrared curvature floor of Section 10:

$$\kappa_r^{\text{sup}}(\mathcal{C}_{IR}^{g.i.}) = \sup_{z \in \mathcal{C}_{IR}^{g.i.}} \kappa_r(z).$$

Definition 7.2 (Refinement preorder). For $x, y \in Z$, define

$$x \preceq_r y \iff \Delta_r(y) \leq \Delta_r(x) \text{ componentwise.}$$

Define

$$x \sim_r y \iff x \preceq_r y \text{ and } y \preceq_r x.$$

Proposition 7.3. *The relation $\preceq_r$ is a preorder, and*

$$x \sim_r y \iff \Delta_r(x) = \Delta_r(y).$$

Proof. Immediate from the componentwise order on $[0, \infty)^5$. □

Definition 7.4 (Global trivialization). A class Z is *globally trivialized* at scale r if there exists a global chart, section, or normalization on Z such that

$$\Delta_r(z) = 0 \quad \forall z \in Z.$$

Proposition 7.5 (Global trivialization collapses the quotient). *If Z is globally trivialized at scale r , then $Z/\sim_r$ has a single class.*

Proof. If $\Delta_r(z) = 0$ for all z , then $\Delta_r(x) = \Delta_r(y)$ for all x, y , hence $x \sim_r y$. $\square$

Theorem 7.6 (Persistent obstruction prevents global trivialization). *If*

$$\inf_{z \in Z} \|\Delta_r(z)\|_\infty > 0,$$

then Z is not globally trivialized at scale r .

Proof. Global trivialization would imply $\Delta_r(z) = 0$ for all z , hence $\|\Delta_r(z)\|_\infty = 0$ for all z , contradicting the positive infimum. $\square$

Theorem 7.7 (Non-constant obstruction prevents collapse). *If Δ_r is not constant on Z , then $Z/\sim_r$ has at least two distinct classes.*

Proof. If Δ_r is not constant, choose $x, y \in Z$ with $\Delta_r(x) \neq \Delta_r(y)$. Then $x \not\sim_r y$. $\square$

Definition 7.8 (Deployment path). *A deployment path at scale r is a chain*

$$z_0 \preceq_r z_1 \preceq_r \cdots \preceq_r z_n.$$

8 Explicit binary model

8.1 Observer-protocol state

The observer-protocol state $\pi = (\alpha, \beta, p)$ summarizes a binary system–instrument–environment measurement scheme. The system carries a latent variable $S \in \{0, 1\}$ with prevalence

$$p := P(S = 1) \in [0, 1],$$

and the instrument produces a readout $M \in \{0, 1\}$ with false positive and false negative rates

$$P(M = 1 \mid S = 0) = \alpha, \quad P(M = 0 \mid S = 1) = \beta.$$

The environment is a register $E \in \{0, 1\}$ independent of (S, M) with $P(E = 1) = \vartheta$, whose influence is encoded in the instrument rates; the observer’s semantic content is the triple (α, β, p) characterizing the instrument. Formally, $X_\pi := X_S \times X_I \times X_E = \{0, 1\}^3$ with the counting reference measure λ_π , the joint density ρ_π given by the law above, the semantic target space $S := [0, 1]^3$, and the semantic map

$$\tau_\pi : X_\pi \rightarrow S, \quad \tau_\pi(s, m, e) := (\alpha, \beta, p),$$

so that the semantic law is the Dirac mass $\nu_\pi = \delta_{(\alpha, \beta, p)}$.

Fix $\eta > 0$ and define the admissible domain

$$D_\eta := \{(\alpha, \beta, p) \in [0, 1]^3 : \alpha + \beta \leq 1 - \eta\}.$$

Define the semantic queries

$$q_0(\alpha, \beta, p) := \alpha, \quad q_1(\alpha, \beta, p) := 1 - \beta, \quad q_*(\alpha, \beta, p) := \alpha(1 - p) + (1 - \beta)p,$$

so that

$$P_\pi(q_0) = \alpha, \quad P_\pi(q_1) = 1 - \beta, \quad P_\pi(q_*) = \alpha(1 - p) + (1 - \beta)p.$$

All three queries are bounded on $S = [0, 1]^3$, and $\|q_0\|_\infty = \|q_1\|_\infty = \|q_*\|_\infty = 1$. Hence

$$\Phi(\pi) = (\alpha, 1 - \beta, \theta(\alpha, \beta, p)),$$

where

$$\theta(\alpha, \beta, p) := \alpha(1 - p) + (1 - \beta)p = \alpha + p(1 - \alpha - \beta).$$

Theorem 8.1 (Exact identity in the binary model). *If $\pi, \pi' \in D_\eta$ and*

$$\Phi(\pi) = \Phi(\pi'),$$

then $\pi = \pi'$.

Proof. Equality of the first two coordinates yields $\alpha = \alpha'$ and $\beta = \beta'$. Equality of the third gives

$$(1 - \alpha - \beta)p = (1 - \alpha - \beta)p'.$$

Since $1 - \alpha - \beta \geq \eta > 0$, one obtains $p = p'$. □

8.2 Agreement window

Define the protocol pseudometric (in fact a metric) and the semantic discrepancy

$$d_{\text{prot}}(\pi, \pi') := \max\{|\alpha - \alpha'|, |\beta - \beta'|, |p - p'|\},$$

and

$$d_{\text{sem}}(\Phi(\pi), \Phi(\pi')) := \max\{|\alpha - \alpha'|, |\beta - \beta'|, |\theta(\alpha, \beta, p) - \theta(\alpha', \beta', p')|\}.$$

Throughout this section, agreement is measured by d_{sem} and the agreement radius is that of Definition 4.3, taken with the closed protocol balls of Convention 5:

$$R_\varepsilon(\pi) := \sup\{r > 0 : B_r^{\text{prot}}(\pi) \subseteq A_\varepsilon(\pi)\}, \quad A_\varepsilon(\pi) := \{\pi' \in D_\eta : d_{\text{sem}}(\Phi(\pi), \Phi(\pi')) \leq \varepsilon\},$$

where $B_r^{\text{prot}}(\pi) := \{\pi' \in D_\eta : d_{\text{prot}}(\pi, \pi') \leq r\}$. Since all queries of the model have sup-norm 1, the normalized agreement modulus of Definition 6.1 satisfies

$$E_\pi(r) = \frac{1}{2} \sup_{\pi' \in B_r^{\text{prot}}(\pi)} d_{\text{sem}}(\Phi(\pi), \Phi(\pi')),$$

so Definition 6.4 yields the same radius with threshold $\varepsilon/2$ in place of ε ; the statements below are written in d_{sem} -units.

Proposition 8.2 (Local Lipschitz estimate). *For all $\pi, \pi' \in D_\eta$,*

$$d_{\text{sem}}(\Phi(\pi), \Phi(\pi')) \leq 2 d_{\text{prot}}(\pi, \pi').$$

Proof. The first two coordinates contribute at most $d_{\text{prot}}(\pi, \pi')$. For the third, the gradient of θ on the convex domain D_η satisfies

$$\frac{\partial \theta}{\partial \alpha} = 1 - p, \quad \frac{\partial \theta}{\partial \beta} = -p, \quad \frac{\partial \theta}{\partial p} = 1 - \alpha - \beta.$$

Since $|1 - p| + |p| = 1$ on $[0, 1]$ and $|1 - \alpha - \beta| \leq 1$, one has

$$\left| \frac{\partial \theta}{\partial \alpha} \right| + \left| \frac{\partial \theta}{\partial \beta} \right| + \left| \frac{\partial \theta}{\partial p} \right| \leq 2.$$

By the mean-value theorem on the convex set D_η ,

$$|\theta(\pi) - \theta(\pi')| \leq 2 d_{\text{prot}}(\pi, \pi').$$

Taking the maximum yields the claim. $\square$

Definition 8.3 (Admissible local radius). Define

$$\text{radm}(\pi) := \sup\{r > 0 : B_r^{\text{prot}}(\pi) \subseteq D_\eta\}.$$

Corollary 8.4 (Lower agreement bound). *For every $\pi \in D_\eta$,*

$$R_\varepsilon(\pi) \geq \min\left(\text{radm}(\pi), \frac{\varepsilon}{2}\right).$$

Proof. For every $0 < s < \min(\text{radm}(\pi), \varepsilon/2)$ one has $B_s^{\text{prot}}(\pi) \subseteq D_\eta$ and, by Proposition 8.2,

$$d_{\text{sem}}(\Phi(\pi), \Phi(\pi')) \leq 2 d_{\text{prot}}(\pi, \pi') \leq 2s < \varepsilon$$

for every $\pi' \in B_s^{\text{prot}}(\pi)$. Hence $B_s^{\text{prot}}(\pi) \subseteq A_\varepsilon(\pi)$ for all such s , and $R_\varepsilon(\pi) \geq \min(\text{radm}(\pi), \varepsilon/2)$. $\square$

8.3 Control words and realized path residue

Define concrete transformations

$$\gamma_u(\alpha, \beta, p) := (\alpha + u, \beta, p), \quad \delta_v(\alpha, \beta, p) := (\alpha, \beta, p + v\alpha),$$

whenever the resulting states remain in D_η .

To instantiate the protocol framework of Section 5, let Γ denote the group generated by the transformations γ_u, δ_v ($u, v \in \mathbb{R}$) acting affinely on $[0, 1]^3$, and set

$$Q_\pi := Q := \left\{ q \circ w : q \in \{q_0, q_1, q_*\}, w \in \Gamma \right\}$$

for every $\pi \in D_\eta$, where each element $q \circ w$ is extended by 0 outside its domain of definition; then $Q \subseteq L^\infty(S)$ and every element of Q has sup-norm at most 1. For an admissible transformation γ_w at π (i.e. $\gamma_w \pi \in D_\eta$), the associated protocol morphism $(K_{\gamma_w}, G_{\gamma_w}, r_{\gamma_w}) : \pi \rightarrow \gamma_w \pi$ is defined by

$$G_{\gamma_w}(q) := q \circ \gamma_w,$$

together with the deterministic state transport K_{γ_w} induced by γ_w on the Dirac laws. Then $G_{\gamma_w}(Q) \subseteq Q$, the compatibility condition

$$P_{\gamma_w \pi}(q) = q(\gamma_w \pi) = (q \circ \gamma_w)(\pi) = P_\pi(G_{\gamma_w} q)$$

holds for every $q \in Q$ by construction, and, since γ_w maps its admissible subdomain into $[0, 1]^3$,

$$L_{\gamma_w} = \sup_{q \in Q} \frac{1 + \|q \circ \gamma_w\|_\infty}{1 + \|q\|_\infty} \leq 1 < \infty.$$

Thus every admissible transformation is a genuine protocol morphism of Definition 5.4. Its realized semantic transport is defined by

$$T_\gamma^{\text{real}}(\Phi(\pi)) := \Phi(\gamma\pi)$$

for every concrete protocol transformation γ generated by γ_u, δ_v , their inverses when defined, and their compositions; by Lemma 5.7 this coincides with the abstract transport $T_\gamma(\Phi_\pi) = \Phi_{\gamma\pi}$ of Definition 5.6. For a concrete control word w , define its realized residue (in profile units) by

$$\text{Res}_{\text{obs}}(w; \pi) := d_{\text{sem}}(\Phi(w\pi), \Phi(\pi)).$$

With the full frame Q , the normalized realized residue of Definition 5.12 for the commutator word below equals the orbit-amplified quantity $\sup_{q \in Q} |q(w\pi) - q(\pi)| / (1 + \|q\|_\infty)$, which is bounded below by $(1 - \beta)|uv|/2 \geq \eta|uv|/2$; the profile-unit version used below is bounded below by $(1 - \alpha - \beta)|uv| \geq \eta|uv|$ (Corollary 8.6).

Proposition 8.5 (Explicit commutator residue). *Define the commutator word*

$$c_{u,v} := \delta_v^{-1} \circ \gamma_u^{-1} \circ \delta_v \circ \gamma_u.$$

Then

$$c_{u,v}(\alpha, \beta, p) = (\alpha, \beta, p + uv).$$

Proof. Direct calculation:

$$(\alpha, \beta, p) \xrightarrow{\gamma_u} (\alpha + u, \beta, p) \xrightarrow{\delta_v} (\alpha + u, \beta, p + v(\alpha + u)) \xrightarrow{\gamma_u^{-1}} (\alpha, \beta, p + v(\alpha + u)) \xrightarrow{\delta_v^{-1}} (\alpha, \beta, p + uv).$$

□

Corollary 8.6 (Residue of the commutator word). *For $\pi = (\alpha, \beta, p) \in D_\eta$,*

$$\text{Res}_{\text{obs}}(c_{u,v}; \pi) = (1 - \alpha - \beta)|uv| \geq \eta|uv|.$$

In particular, for $u, v \geq 0$,

$$\text{Res}_{\text{obs}}(c_{u,v}; \pi) = (1 - \alpha - \beta)uv \geq \eta uv.$$

Proof. The endpoint is $c_{u,v}\pi = (\alpha, \beta, p + uv)$. The first two semantic coordinates remain unchanged. The third changes by

$$\theta(\alpha, \beta, p + uv) - \theta(\alpha, \beta, p) = (1 - \alpha - \beta)uv.$$

The residue is the absolute value of this quantity. Use $1 - \alpha - \beta \geq \eta$. □

Definition 8.7 (Balanced-word admissibility set and radius). Define

$$S_\pi := \{r > 0 : c_{r/2, r/2} \text{ is defined and all intermediate states of the word lie in } B_r^{\text{prot}}(\pi)\},$$

and

$$r_{\text{word}}(\pi) := \sup S_\pi.$$

For every interior state the admissible set is non-trivial: if $\alpha + \beta < 1 - \eta$ and $0 < p < 1$, then

$$r_{\text{word}}(\pi) \geq \min\left(4, 2(1 - \eta - \alpha - \beta), \sqrt{\alpha^2 + 4(1 - p)} - \alpha\right) > 0,$$

since the constraints $\alpha + t/2 + \beta \leq 1 - \eta$, $p + (t/2)(\alpha + t/2) \leq 1$, and $\max(t/2, (t/2)(\alpha + t/2), t^2/4) \leq t$ all hold for sufficiently small $t > 0$ (the first gives $t/2 \leq 1 - \eta - \alpha - \beta$, hence $\alpha + t/2 \leq 1 - \eta - \beta < 1 \leq 2$ and $(t/2)(\alpha + t/2) \leq t$; and $t^2/4 \leq t$ for $t \leq 4$).

Lemma 8.8 (Initial-interval property of the admissible set). *For every $\pi \in D_\eta$, the set S_π is downward closed:*

$$r \in S_\pi, 0 < s < r \implies s \in S_\pi.$$

Consequently, if $r_{\text{word}}(\pi) > a$, then there exists $r \in S_\pi$ with $a < r < r_{\text{word}}(\pi)$.

Proof. Let $\pi = (\alpha, \beta, p)$. The intermediate states of $c_{u,v}$ are

$$(\alpha + u, \beta, p), \quad (\alpha + u, \beta, p + v(\alpha + u)), \quad (\alpha, \beta, p + v(\alpha + u)),$$

and the endpoint is

$$(\alpha, \beta, p + uv).$$

For the balanced word $u = v = t/2$, each coordinate increment is a nondecreasing function of $t \geq 0$:

$$u = \frac{t}{2}, \quad v(\alpha + u) = \frac{t}{2} \left(\alpha + \frac{t}{2} \right), \quad uv = \frac{t^2}{4}.$$

Hence, if the admissibility inequalities defining D_η and the ball-inclusion inequalities defining $B_t^{\text{prot}}(\pi)$ hold at $t = r$, then they also hold at every $0 < s < r$, because the corresponding increments at s are bounded above by those at r . Thus S_π is downward closed.

If $r_{\text{word}}(\pi) > a$, then by definition of supremum there exists $r \in S_\pi$ with $a < r \leq r_{\text{word}}(\pi)$. Since S_π is downward closed, any $r' \in (a, r)$ also lies in S_π . $\square$

Theorem 8.9 (Finite upper agreement bound in the binary model). *If*

$$r_{\text{word}}(\pi) > \sqrt{\frac{4\varepsilon}{\eta}},$$

then

$$R_\varepsilon(\pi) \leq \sqrt{\frac{4\varepsilon}{\eta}}.$$

Proof. Assume by contradiction that

$$R_\varepsilon(\pi) > \sqrt{\frac{4\varepsilon}{\eta}}.$$

By the previous lemma, choose

$$r \in S_\pi$$

such that

$$\sqrt{\frac{4\varepsilon}{\eta}} < r < \min\{R_\varepsilon(\pi), r_{\text{word}}(\pi)\}.$$

Since $r < R_\varepsilon(\pi)$ and the agreement radius is defined by ball inclusion, there exists $s \in (r, R_\varepsilon(\pi))$ with $B_s^{\text{prot}}(\pi) \subseteq A_\varepsilon(\pi)$; the balls are nested, so $B_r^{\text{prot}}(\pi) \subseteq A_\varepsilon(\pi)$, i.e. every state in $B_r^{\text{prot}}(\pi)$ is ε -close semantically to π . Because $r \in S_\pi$, all intermediate states and the endpoint of the balanced commutator word $c_{r/2, r/2}$ lie in the ball. Therefore

$$\text{Res}_{\text{obs}}(c_{r/2, r/2}; \pi) \leq \varepsilon.$$

But the previous corollary yields

$$\text{Res}_{\text{obs}}(c_{r/2, r/2}; \pi) \geq \frac{\eta r^2}{4} > \varepsilon,$$

contradiction. $\square$

Corollary 8.10 (Two-sided agreement window). *If*

$$r_{\text{word}}(\pi) > \sqrt{\frac{4\varepsilon}{\eta}},$$

then

$$\min\left(\text{radm}(\pi), \frac{\varepsilon}{2}\right) \leq R_\varepsilon(\pi) \leq \sqrt{\frac{4\varepsilon}{\eta}}.$$

Proof. Combine the lower and upper bounds. $\square$

The binary model also exhibits non-trivial *protocol holonomy* in the sense of Definition 5.9. For $\pi = (\alpha, \beta, p)$ in the interior of D_η (i.e. $\alpha + \beta < 1 - \eta$) and $v \neq 0$, consider the loop

$$\ell = \gamma_\alpha \circ \delta_v \circ \gamma_{-\alpha}$$

at π (for $\alpha = 0$ take simply $\ell = \delta_v$). Its action on states is

$$\ell(\alpha', \beta, p') = (\alpha', \beta, p' + v(\alpha' - \alpha)),$$

so $\ell\pi = \pi$ while ℓ is not the identity on D_η ; its composite pullback satisfies $G_\ell(q) = q \circ \ell \neq q$ for $q = q_*$. Choosing the bounded functional u on Q with $u(q_* \circ \ell) = 1 + \|q_* \circ \ell\|_\infty$ and $u(q_*) = -1 - \|q_*\|_\infty$ (both values admissible, since $\|u\|_\pi \leq 1$), one obtains

$$\text{Hol}_{\text{abs}}(\ell; \pi) \geq \frac{|u(q_* \circ \ell) - u(q_*)|}{1 + \|q_*\|_\infty} = \frac{(1 + \|q_* \circ \ell\|_\infty) + (1 + \|q_*\|_\infty)}{1 + \|q_*\|_\infty} \geq 2.$$

Since the general bound $\text{Hol}_{\text{abs}}(\ell; \pi) \leq L_\ell + 1 \leq 2$ holds (as $L_\ell \leq 1$), one has

$$\text{Hol}_{\text{abs}}(\ell; \pi) = 2.$$

Thus the binary model carries non-trivial abstract observational holonomy, localized in the ball $B_\alpha^{\text{prot}}(\pi)$.

9 Non-closure in the binary model

We now instantiate the obstruction vector of Section 7 in the binary model, taking $Z := D_\eta$ with the protocol-morphism structure of Section 8.3. The agency base is the one-object category A whose single object is D_η and whose morphisms are the admissible words $w \in \Gamma$, endowed with the transport data of Section 8.3 ($X_a := D_\eta$ with the counting measure, K_w the deterministic transport $\pi \mapsto w\pi$, and $G_w(q) := q \circ w$); every word is then a loop at the unique object. The certificate class is trivial: $\Sigma_{\text{cert}} := \{\text{id}\}$ with $\text{cost}(\text{id}) := 0$ and $\text{err}(\text{id}, \pi) := d_{\text{sem}}(\Phi(\pi), \Phi(\pi)) = 0$, reflecting that the binary model is solved exactly.

Lemma 9.1 (Base-query atlas discrepancy of a word). *For every admissible word w and every $\pi = (\alpha, \beta, p) \in D_\eta$,*

$$d_a^{\text{base}}(\Gamma_w(\xi_\pi), \xi_\pi) = \frac{1}{2} d_{\text{sem}}(\Phi(w\pi), \Phi(\pi)).$$

Proof. The base query family of the binary chart is $\{q_0, q_1, q_*\}$, all of sup-norm 1; by Definition 2.6,

$$d_a^{\text{base}}(\Gamma_w(\xi_\pi), \xi_\pi) = \frac{1}{2} \max\left\{|q_0(w\pi) - q_0(\pi)|, |q_1(w\pi) - q_1(\pi)|, |q_*(w\pi) - q_*(\pi)|\right\} = \frac{1}{2} d_{\text{sem}}(\Phi(w\pi), \Phi(\pi)).$$

$\square$

Lemma 9.2 (Reachability). *Let $\pi = (\alpha, \beta, p) \in D_\eta$.*

(i) *Every admissible word maps π into the rectangle*

$$R_\pi := [0, 1 - \eta - \beta] \times \{\beta\} \times [0, 1].$$

(ii) *If $\alpha > 0$, then every point of R_π is reachable from π by an admissible word.*

(iii) *If $\alpha = 0$, then every point of $\{0\} \times \{\beta\} \times [0, 1]$ is reachable from π .*

Proof. (i) Every word preserves the β -coordinate and keeps the resulting state in D_η , whence $\alpha(w\pi) \leq 1 - \eta - \beta$ and $p(w\pi) \in [0, 1]$.

(ii) Given a target $(\alpha', \beta, p') \in R_\pi$: if $\alpha' > 0$, apply first $\gamma_{\alpha' - \alpha}$ and then $\delta_{(p' - p)/\alpha'}$; the intermediate and final states are

$$(\alpha', \beta, p) \xrightarrow{\delta_{(p' - p)/\alpha'}} (\alpha', \beta, p'),$$

with $\alpha' \leq 1 - \eta - \beta$, so all states lie in D_η . If $\alpha' = 0$ (and $\alpha > 0$), apply first $\delta_{(p' - p)/\alpha}$ and then $\gamma_{-\alpha}$: $p \mapsto p'$, then $\alpha \mapsto 0$, with p' unchanged by γ .

(iii) For $\alpha = 0$ and a target $(0, \beta, p')$: choose $u \neq 0$ with $0 < u \leq 1 - \eta - \beta$ and apply γ_u , then $\delta_{(p' - p)/u}$, then γ_{-u} ; the states

$$(u, \beta, p) \xrightarrow{\delta_{(p' - p)/u}} (u, \beta, p') \xrightarrow{\gamma_{-u}} (0, \beta, p')$$

all lie in D_η . □

Theorem 9.3 (Obstruction vector of the binary model). *Let $\pi = (\alpha, \beta, p) \in D_\eta$. Then:*

(i) $\kappa_r(\pi) = 0$ for every $r > 0$;

(ii) $\text{Hol}_r^{\text{atlas}}(\pi) = \frac{1}{2} \max(\max(\alpha, 1 - \eta - \beta - \alpha), \max(\theta(\pi), 1 - \beta - \theta(\pi))) \geq \eta/4$ for every $r > 0$;

(iii) $\text{Hol}_r^{\text{prot}}(\pi) = 2$ whenever $r \geq \alpha$ and $\alpha + \beta < 1 - \eta$, and $0 \leq \text{Hol}_r^{\text{prot}}(\pi) \leq 2$ for all $r \geq 0$;

(iv) if $r < r_{\text{word}}(\pi)$, then $\eta r^2/8 \leq g_r(\pi) \leq 1$ and $\eta r^2/8 \leq c_r(\pi) \leq 2r$.

Consequently

$$\Delta_r(\pi) = \left(0, \text{Hol}_r^{\text{atlas}}(\pi), \text{Hol}_r^{\text{prot}}(\pi), g_r(\pi), c_r(\pi)\right)$$

with the bounds above.

Proof. (i) With the trivial certificate class, $\text{err}(\text{id}, \pi) = 0$ and $\text{cost}(\text{id}) = 0 \leq r$, so $\kappa_r(\pi) = 0$.

(ii) By Lemmas 9.1 and 9.2,

$$\text{Hol}_r^{\text{atlas}}(\pi) = \frac{1}{2} \sup_{\pi' \in R_\pi} d_{\text{sem}}(\Phi(\pi'), \Phi(\pi)).$$

On the reachable rectangle R_π the α -coordinate ranges over $[0, 1 - \eta - \beta]$ and the profile coordinate θ ranges over $[0, 1 - \beta]$ (its extremes are attained at $(0, \beta, 0)$ and $(0, \beta, 1)$, both reachable). Hence

$$\sup_{\pi' \in R_\pi} d_{\text{sem}}(\Phi(\pi'), \Phi(\pi)) = \max\left(\max(\alpha, 1 - \eta - \beta - \alpha), \max(\theta(\pi), 1 - \beta - \theta(\pi))\right),$$

and, since $\theta(\pi) + (1 - \beta - \theta(\pi)) = 1 - \beta \geq \eta$, the second maximum is at least $\eta/2$; therefore $\text{Hol}_r^{\text{atlas}}(\pi) \geq \eta/4$.

(iii) The loop $\ell = \gamma_\alpha \circ \delta_v \circ \gamma_{-\alpha}$ (or $\ell = \delta_v$ if $\alpha = 0$) of Section 8.3, for interior states $\alpha + \beta < 1 - \eta$, satisfies $\text{Hol}_{\text{abs}}(\ell; \pi) = 2$ and is contained in $B_r^{\text{prot}}(\pi)$ exactly when $r \geq \alpha$ (its intermediate state is $(0, \beta, p)$, at distance α from π). For every loop, $\text{Hol}_{\text{abs}}(\ell; \pi) \leq L_\ell + 1 \leq 2$ since each morphism of the model has $L \leq 1$. Hence $\text{Hol}_r^{\text{prot}}(\pi) = 2$ for $r \geq \alpha$, and $0 \leq \text{Hol}_r^{\text{prot}}(\pi) \leq 2$ in general.

(iv) For $r < r_{\text{word}}(\pi)$ the balanced commutator word $c_{r/2, r/2}$ is admissible. Its normalized realized residue satisfies, by the paragraph following Section 8.3 and Corollary 8.6,

$$\text{Res}_{\text{obs}}(p_{c_{r/2, r/2}}; \pi) \geq \frac{(1 - \beta) r^2}{8} \geq \frac{\eta r^2}{8},$$

whence $g_r(\pi) \geq \eta r^2/8$; the upper bound $g_r(\pi) \leq 1$ holds because each normalized discrepancy term is at most $2\|q\|_\infty/(1 + \|q\|_\infty) \leq 1$. For the compression obstruction, any center $o^* \in C_r(\pi)$ satisfies, with $e := c_{r/2, r/2} \pi \in C_r(\pi)$ (note $d_{\text{prot}}(\pi, e) = r^2/4 \leq r$),

$$\sup_{o \in C_r(\pi)} d_{\text{sem}}(\Phi(o), \Phi(o^*)) \geq \frac{1}{2} \left[d_{\text{sem}}(\Phi(\pi), \Phi(o^*)) + d_{\text{sem}}(\Phi(o^*), \Phi(e)) \right] \geq \frac{1}{2} d_{\text{sem}}(\Phi(\pi), \Phi(e)) \geq \frac{(1 - \alpha - \beta) r^2}{8} \geq$$

so $c_r(\pi) \geq \eta r^2/8$; and with the center $o^* = \pi$, Proposition 8.2 gives $\sup_{o \in C_r(\pi)} d_{\text{sem}}(\Phi(o), \Phi(\pi)) \leq 2r$, so $c_r(\pi) \leq 2r$. $\square$

Corollary 9.4 (Persistent non-closure prevents trivialization in the binary model). *For every $\pi \in D_\eta$ and every $r > 0$,*

$$\|\Delta_r(\pi)\|_\infty \geq \text{Hol}_r^{\text{atlas}}(\pi) \geq \frac{\eta}{4} > 0.$$

Hence

$$\inf_{\pi \in D_\eta} \|\Delta_r(\pi)\|_\infty \geq \frac{\eta}{4} > 0,$$

and by Theorem 7.6 the binary model is not globally trivialized at any scale r ; its non-closure is a persistent obstruction.

Proof. The second coordinate of $\Delta_r(\pi)$ is $\text{Hol}_r^{\text{atlas}}(\pi) \geq \eta/4$ by Theorem 9.3(ii). The rest follows from Theorem 7.6. $\square$

Theorem 9.5 (Order by non-closure in the binary model). *The obstruction vector Δ_r is non-constant on D_η : for $\eta = 0.1$ one has*

$$\text{Hol}_r^{\text{atlas}}(0.1, 0.2, 0.3) = 0.3 \quad \text{and} \quad \text{Hol}_r^{\text{atlas}}(0.1, 0.5, 0.3) = 0.15.$$

Consequently, by Theorem 7.7, the quotient $D_\eta/\sim_r$ has at least two distinct classes for every $r > 0$. Moreover, along the family $\pi_\beta := (\alpha, \beta, p)$ with fixed α and $0 < p < 1$, the projected preorder

$$x \preceq_r^{\text{atlas}} y \iff \text{Hol}_r^{\text{atlas}}(y) \leq \text{Hol}_r^{\text{atlas}}(x)$$

is total: $\pi_\beta \preceq_r^{\text{atlas}} \pi_{\beta'} \iff \beta' \geq \beta$, and any chain $\beta_1 \leq \beta_2 \leq \dots$ yields a deployment path

$$\pi_{\beta_1} \preceq_r^{\text{atlas}} \pi_{\beta_2} \preceq_r^{\text{atlas}} \dots,$$

strict whenever the β 's are distinct.

Proof. By Theorem 9.3(ii), $\text{Hol}_r^{\text{atlas}}(\alpha, \beta, p) = \frac{1}{2} \max(\max(\alpha, 1 - \eta - \beta - \alpha), \max(\theta, 1 - \beta - \theta))$, which gives the two displayed values ($\max(0.6, 0.49) = 0.6$ at $(0.1, 0.2, 0.3)$ and $\max(0.3, 0.28) = 0.3$ at $(0.1, 0.5, 0.3)$, halved), and, for $0 < p < 1$, every coordinate appearing in the maximum is strictly decreasing in β (namely $1 - \eta - \beta - \alpha$, $\theta = \alpha + p(1 - \alpha - \beta)$, and $1 - \beta - \theta = (1 - p)(1 - \alpha - \beta)$); distinct β 's therefore give distinct obstruction vectors, so the corresponding states are not $\sim_r$ -equivalent (Theorem 7.7). $\square$

10 Conditional Yang–Mills reduction

10.1 Spectral implication of exponential clustering

Let $(\mathcal{H}_{\text{phys}}, \Omega, H)$ be a Euclidean-OS reconstructed gauge-invariant physical Hilbert space [3, 4, 7], where H is self-adjoint, $H \geq 0$, $H\Omega = 0$, and Ω is a normalized vector which is the unique vacuum state, so that the spectral projector $E(\{0\})$ of H onto $\{0\}$ satisfies $E(\{0\}) = |\Omega\rangle\langle\Omega|$. For a local gauge-invariant observable O , define

$$\tilde{O} := O - \langle\Omega, O\Omega\rangle,$$

and

$$C_O(t) := \langle\Omega, \tilde{O} e^{-tH} \tilde{O} \Omega\rangle.$$

Definition 10.1 (Spectrally separating family). A family M of local gauge-invariant observables is *spectrally separating* if

$$\overline{\text{span}}\{\tilde{O} \Omega : O \in M\} = \mathcal{H}_{\text{phys}} \ominus \mathbb{R}\Omega,$$

where the span is the closed linear span.

Proposition 10.2 (Exponential clustering implies spectral mass gap). *Suppose there exists $c > 0$ such that for a spectrally separating family M of local gauge-invariant observables,*

$$C_O(t) \leq A_O e^{-ct} \quad \forall O \in M, \text{ for } t \gg 1.$$

Then

$$\text{Spec}(H) \subseteq \{0\} \cup [c, \infty),$$

hence the theory has mass gap $\Delta \geq c$.

Proof. By the spectral theorem [2],

$$C_O(t) = \int_{[0, \infty)} e^{-tm} d\mu_O(m)$$

for a positive measure μ_O . Since $\tilde{O}\Omega \perp \Omega$ and $E(\{0\}) = |\Omega\rangle\langle\Omega|$, one has $\mu_O(\{0\}) = \|E(\{0\})\tilde{O}\Omega\|^2 = |\langle\Omega, \tilde{O}\Omega\rangle|^2 = 0$. If $\mu_O((0, c)) > 0$, then by continuity from below of the finite measure μ_O there exists $\delta > 0$ with

$$\mu_O([0, c - \delta]) > 0.$$

Hence

$$C_O(t) \geq \mu_O([0, c - \delta]) e^{-(c-\delta)t},$$

contradicting the assumed upper bound $A_O e^{-ct}$ for large t (since $e^{\delta t} \rightarrow \infty$). Therefore $\mu_O((0, c)) = 0$ for all $O \in M$.

Let $E_{(0, c)}$ denote the spectral projector of H on $(0, c)$. Suppose for contradiction that

$$E_{(0, c)} \neq 0$$

on

$$\mathcal{H}_{\text{phys}} \ominus \mathbb{R}\Omega.$$

Then there exists a nonzero vector in $\mathcal{H}_{\text{phys}} \ominus \mathbb{R}\Omega$ with nontrivial projection onto $(0, c)$. Since M is spectrally separating and $E_{(0, c)}$ is continuous,

$$\overline{\text{span}}\{\tilde{O} \Omega : O \in M\} = \mathcal{H}_{\text{phys}} \ominus \mathbb{R}\Omega,$$

so there must exist some $O \in M$ such that

$$E_{(0,c)} \tilde{O} \Omega \neq 0.$$

For this observable,

$$\mu_O((0,c)) = \langle \tilde{O} \Omega, E_{(0,c)} \tilde{O} \Omega \rangle = \|E_{(0,c)} \tilde{O} \Omega\|^2 > 0,$$

contradicting the previous paragraph. Hence

$$E_{(0,c)} = 0$$

on

$$\mathcal{H}_{\text{phys}} \ominus \mathbb{R}\Omega,$$

which is equivalent to

$$\text{Spec}(H) \subseteq \{0\} \cup [c, \infty).$$

Therefore the theory has mass gap $\Delta \geq c$. □

10.2 Audited Yang–Mills interface

Definition 10.3 (Certificate class). Let Σ_{cert} be a class of admissible certificates, models, or protocol representations together with a cost map

$$\text{cost} : \Sigma_{\text{cert}} \rightarrow [0, \infty).$$

Let $\mathcal{C}_{IR}^{g.i.} \subseteq Z$ denote the class of *gauge-invariant infrared semantic states*: the protocol states obtained from finite-volume lattice gauge configurations in the infrared regime together with their finite-bank transfers toward the continuum regime (Hypothesis 10.8 below).

Definition 10.4 (Audited Yang–Mills interface). An *audited Yang–Mills interface* is a tuple

$$\mathcal{S}_{\text{YM}} = (\Theta, \Sigma_r, \Sigma_{\text{cert}}, M_r, \text{err}, \mu, r),$$

where:

1. Θ is a parameter space (a, L, β, m_q, ξ) of lattice spacing, volume, bare couplings, masses, and protocol data;
2. Σ_r is a finite bank of lattice gauge configurations and admissible protocol operations at resource level r ;
3. Σ_{cert} is the admissible certificate class;
4. M_r is a class of gauge-invariant observables;
5. err is a certification error functional

$$\text{err} : \Sigma_{\text{cert}} \times Z \rightarrow [0, \infty);$$

6. μ is an admissibility measure on Σ_{cert} .

Definition 10.5 (Infrared curvature floor). For the infrared gauge-invariant semantic class $\mathcal{C}_{IR}^{g.i.}$, define

$$\kappa_r^{\sup}(\mathcal{C}_{IR}^{g.i.}) := \sup_{o \in \mathcal{C}_{IR}^{g.i.}} \inf_{\sigma \in \Sigma_{\text{cert}} : \text{cost}(\sigma) \leq r} \text{err}(\sigma, o),$$

and

$$\kappa_*^{\sup}(\mathcal{C}_{IR}^{g.i.}) := \liminf_{r \rightarrow \infty} \kappa_r^{\sup}(\mathcal{C}_{IR}^{g.i.}).$$

Definition 10.6 (UV/IR protocol subcategory). Let

$$\mathbf{Prot}_{\text{UV/IR}} \subseteq \mathbf{Prot}$$

be the smallest subcategory generated by five specified families of morphisms:

1. *continuum extrapolation morphisms* $\gamma_{a \rightarrow a/2} : \pi_a \rightarrow \pi_{a/2}$, carrying finite-lattice states of spacing a to the doubled (finer) lattice, with pullback embedding the coarse observables into the fine ones;
2. *volume extrapolation morphisms* $\gamma_{L \rightarrow 2L} : \pi_L \rightarrow \pi_{2L}$, doubling the spatial volume;
3. *renormalization morphisms*: the Wilson-type [5] flow of the bare couplings (a, L, β, m_q) along the renormalization group, with scale-dependent parameters;
4. *observable-choice morphisms*: query pullbacks induced by inclusions of observable algebras along admissible gauge-invariant test functions;
5. *finite-bank transfer morphisms*: the morphisms realizing the transfer modulus ω_r of Hypothesis 10.8 between finite banks and the ideal continuum.

Each of these families consists of protocol morphisms satisfying the semantic compatibility condition of Definition 5.4 by construction.

Definition 10.7 (Nontrivial UV/IR abstract holonomy). We say that $\mathbf{Prot}_{\text{UV/IR}}$ has *nontrivial UV/IR abstract holonomy* if there exist a protocol state $\pi \in \text{Ob}(\mathbf{Prot}_{\text{UV/IR}})$ and a loop ℓ at π in $\mathbf{Prot}_{\text{UV/IR}}$ such that

$$\text{Hol}_{\text{abs}}(\ell; \pi) > 0.$$

Persistence of nontrivial UV/IR abstract holonomy means: there exist $c > 0$ and a resource scale r_0 such that for every $r \geq r_0$ there exist $\pi \in \text{Ob}(\mathbf{Prot}_{\text{UV/IR}})$ and a loop ℓ at π in $\mathbf{Prot}_{\text{UV/IR}}$ with

$$\text{Hol}_{\text{abs}}(\ell; \pi) \geq c \quad \text{and} \quad r_\pi \succeq r,$$

Hypothesis 10.8 (Finite-bank transfer to the ideal continuum). *There exists a transfer modulus $\omega_r \rightarrow 0$ carrying finite-bank protocol semantics to the ideal continuum regime.*

Conjecture 10.9 (Bridge A: abstract holonomy to curvature). *Persistence of nontrivial UV/IR abstract holonomy implies*

$$\mathbf{Prot}_{\text{UV/IR}} \text{ has nontrivial UV/IR abstract holonomy} \implies \kappa_*^{\sup}(\mathcal{C}_{IR}^{g.i.}) > 0.$$

Conjecture 10.10 (Bridge B: curvature to exponential clustering). *A positive infrared curvature floor implies the existence of $c > 0$ and a spectrally separating family M of local gauge-invariant observables such that*

$$C_O(t) \leq A_O e^{-ct} \quad \forall O \in M.$$

Theorem 10.11 (Conditional Yang–Mills reduction theorem). *Assume:*

1. *a reflection-positive audited lattice family and OS reconstruction;*
2. *finite-bank transfer to the ideal continuum;*
3. *persistence of nontrivial UV/IR abstract holonomy;*
4. *Bridge A and Bridge B.*

Then the reconstructed Yang–Mills theory has a strictly positive mass gap [6].

Proof. Assumption 1 provides, via reflection positivity and OS reconstruction, the physical Hilbert space $(\mathcal{H}_{\text{phys}}, \Omega, H)$ of Section 10.1, to which Proposition 10.2 applies. By assumption 3, nontrivial UV/IR abstract holonomy persists along the resource scale, and Bridge A (assumption 4) transfers this persistent defect to the curvature floor:

$$\kappa_*^{\text{sup}}(\mathcal{C}_{IR}^{g.i.}) > 0.$$

Bridge B then yields $c > 0$ and a spectrally separating family M of local gauge-invariant observables with $C_O(t) \leq A_O e^{-ct}$ for all $O \in M$ and $t \gg 1$. Assumption 2 guarantees that these finite-bank estimates transfer to the reconstructed continuum theory through the modulus $\omega_r \rightarrow 0$. Proposition 10.2 now applies and gives

$$\text{Spec}(H) \subseteq \{0\} \cup [c, \infty),$$

so the mass gap satisfies $\Delta \geq c > 0$. □

References

- [1] Saunders Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Graduate Texts in Mathematics 5, Springer, 1998.
- [2] Michael Reed and Barry Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, revised and enlarged edition, Academic Press, 1980.
- [3] Konrad Osterwalder and Robert Schrader, *Axioms for Euclidean Green’s Functions*, Communications in Mathematical Physics 31 (1973), 83–112.
- [4] Konrad Osterwalder and Robert Schrader, *Axioms for Euclidean Green’s Functions II*, Communications in Mathematical Physics 42 (1975), 281–305.
- [5] Kenneth G. Wilson, *Confinement of Quarks*, Physical Review D 10 (1974), 2445–2459.
- [6] Arthur Jaffe and Edward Witten, *Quantum Yang–Mills Theory*, The Millennium Prize Problems, Clay Mathematics Institute, 2000.
- [7] James Glimm and Arthur Jaffe, *Quantum Physics: A Functional Integral Point of View*, 2nd ed., Springer, 1987.
- [8] Oscar Riveros, *General Epistemic Geometry*, GitHub repository, <https://github.com/maxtuno/General-Epistemic-Geometry>.